COSMOS benchmark methodology

How the numbers on the website were produced, scored, and where they stop being reliable.

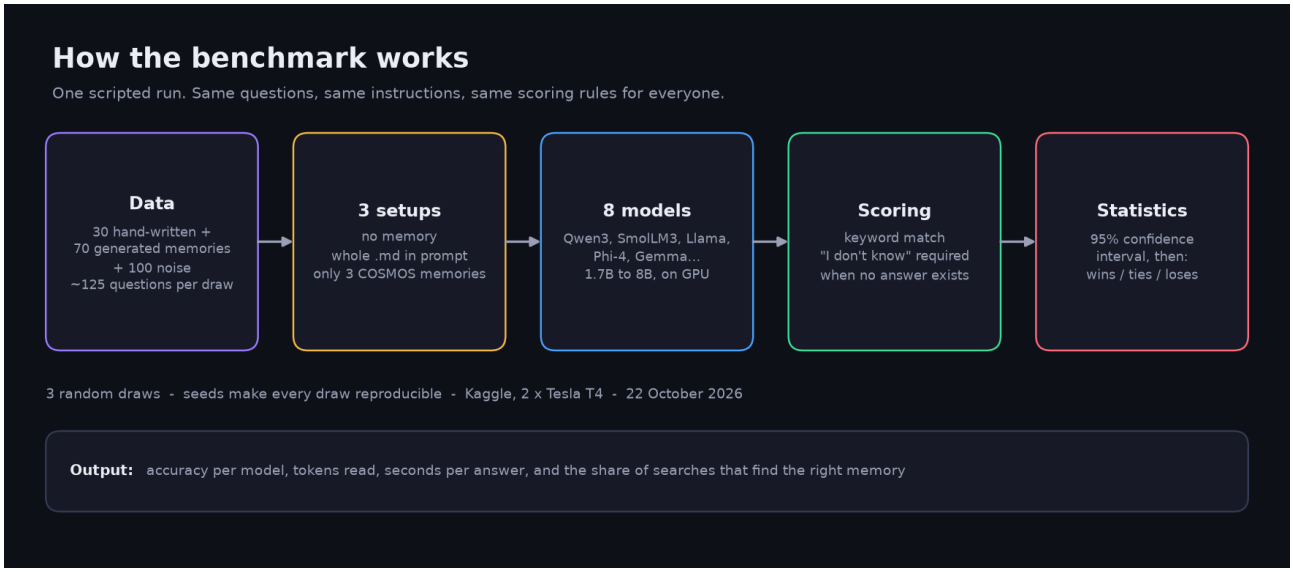
What	Detail
Question asked	Does an AI assistant answer better, faster and with less to read when it gets only the few memories COSMOS finds, instead of the whole notes file pasted in the prompt?
Compared	No memory, the full notes file in the prompt ("full .md"), and COSMOS (3 memories).
Models	8 open models, 1.7 to 8 billion parameters, run locally on GPU.
Size	About 200 stored memories; 298 questions with an answer per model and per setup, plus about 77 questions with no answer in memory (reported separately).
Run	Kaggle, 2 x Tesla T4 GPUs, 22 October 2026. Results published in November 2026.
Code and data	Everything ships with the project: `benchmark/` (code) and `benchmark-cosmos/` (raw results, every answer of every model).

This document describes what the code actually does. It also lists the limits of the test in section 9. COSMOS wins on 4 models, ties on 2 and loses on 2 against the full file: the test is not tuned to show wins everywhere.

1. Three measurements

The benchmark has three parts. Only part B uses an AI model to answer.

Part	Name	What it measures	Needs a language model?
A	Search alone	Does the search find the right memory? How fast, as the memory grows from 150 to 1,700 entries.	No
B	Answers with an AI	Does a model give the right answer with no memory, the full file, or COSMOS?	Yes (8 models)
C	Polaris vs other search models	Does the fine-tuned search model find memories better than the models it replaces?	No



The path every number takes, from test data to verdict.

2. Test data

Everything is in French, because COSMOS is mainly used in French. The data has three sources.

2.1 Hand-written memories and questions

- **30 memories** written by hand (not generated), in six categories: facts (10), preferences (5), project context (5), decisions (4), tasks (3), open questions (3). They live in `benchmark/dataset.json`.
- **38 questions** written by hand about them: 23 direct (same words as the memory), 5 rephrased (same meaning, different words), 2 about an update (a fact changed, the newest value is expected), 2 about history (the old value is expected), 6 with no answer in memory.

2.2 Generated cases

- For each random draw ("seed"), `benchmark/synth.py` generates **60 cases**, each about a made-up person (invented surname, invented company and pet names, random codes and dates). A model cannot answer from its own knowledge, only from the memory.
- Six case types rotate: pet name, job, city, birthday, safe code, and move (old city, then new city). A move produces two memories, the second one marked as replacing the first; half of them also get a history question.
- Every third case adds a question with no answer (phone number, car color, favorite dish, height, salary).
- The seed fixes every random choice, so a draw can be regenerated identically.

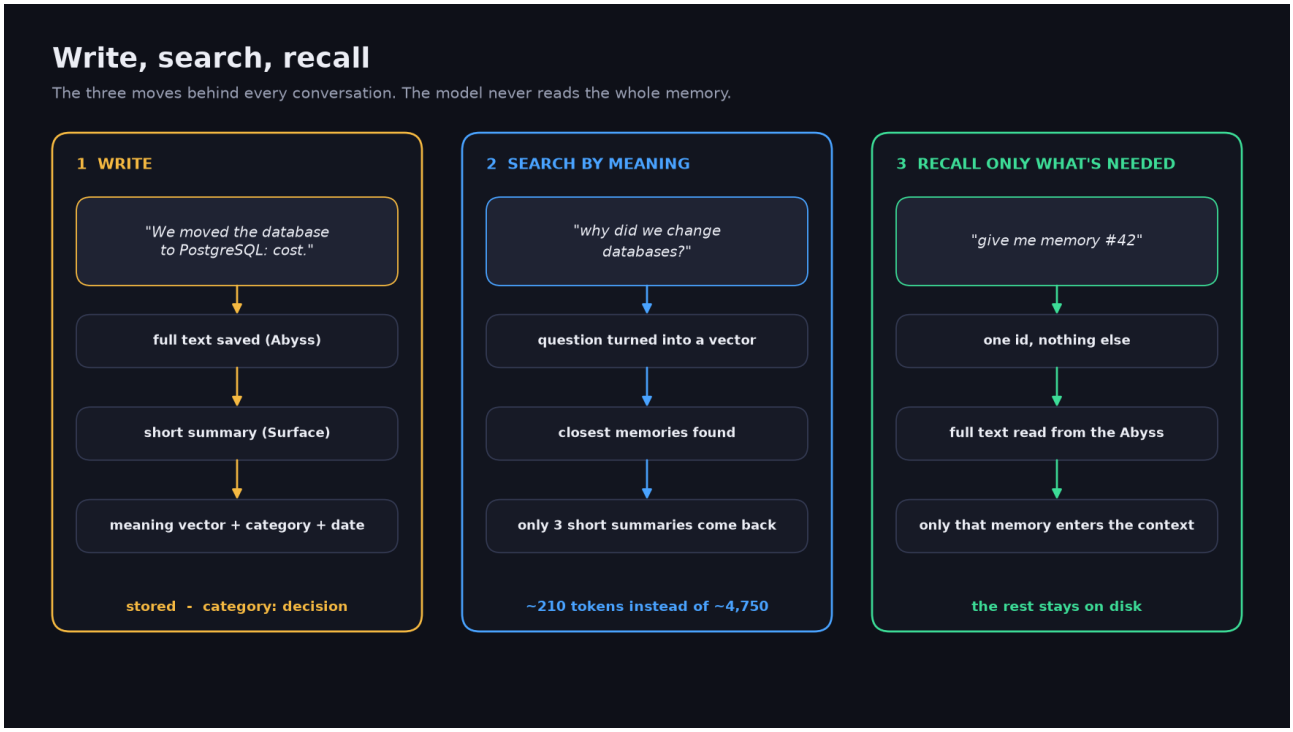
2.3 Noise

- Unrelated memories built from templates ("X lives in Y for N years", "appointment with X on a day at an hour", "X repainted the living room"...), inserted at random positions without changing the order of the real memories.
- For the AI tests, 100 noise memories were added. Total: 30 + 70 generated + 100 noise = **about 200 memories**. The search-only series goes from 50 to 1,600 noise memories (150 to 1,700 memories in total).

	Per draw	3 draws (AI tests)
Memories (with 100 noise)	about 200	same data, 3 different random draws
Questions	about 125 (about 99 with an answer, about 26 without)	375 (298 with an answer, about 77 without)

3. The three setups compared

Every model receives the same question with the same instructions. Only the memory differs.



Background: what COSMOS does. The "cosmos" setup below uses step 2 (search by meaning) to pick the 3 memories.

Setup	What the model reads
none	"(no memory)". The floor: it shows the model cannot answer by itself.
full .md	All memories pasted in the prompt, as a list, numbered in the order they were written.
cosmos	Only the 3 memories COSMOS finds for the question, in chronological order. Each memory shown is the full text. If COSMOS knows a memory was replaced by a newer one, the line says so.



The two memory setups side by side. Each block stands for about 2 memories (illustration).

Each line has the form `- [memory no. 12] text`, or `- [memory no. 12, replaced by a newer one] text`. A higher number means a more recent memory.

An asymmetry you should know about. Only COSMOS can mark a memory as replaced, because it detects versions of the same fact. The full .md file has the numbers (so a model can reason that a higher number is newer) but no such mark. This favors COSMOS on update questions. It is part of what COSMOS does, but it is

not a like-for-like comparison on that category.

Instructions given to the model (system prompt, in French; the English translation follows)

```
Tu es un assistant qui connait l'utilisateur grace a une memoire. Reponds en francais, en
une phrase courte, uniquement a partir de la memoire fournie. Si la reponse n'est pas dans
la memoire, reponds exactement : Je ne sais pas. Chaque souvenir a un numero : plus il est
grand, plus il est recent. Si deux souvenirs se contredisent, le plus recent fait foi, sauf
si la question porte sur le passe.
```

"You are an assistant who knows the user thanks to a memory. Answer in French, in one short sentence, only from the memory provided. If the answer is not in the memory, answer exactly: I don't know. Each memory has a number: the bigger, the more recent. If two memories contradict each other, the most recent one wins, unless the question is about the past."

4. Models and how they were run

Model	Maker	Parameters	Hugging Face id
Qwen3 1.7B	Qwen	1.7B	Qwen/Qwen3-1.7B
SmolLM3 3B	Hugging Face	3.1B	HuggingFaceTB/SmolLM3-3B
Llama 3.2 3B Instruct	Meta	3.2B	unsloth/Llama-3.2-3B-Instruct
Phi-4 mini Instruct	Microsoft	3.8B	microsoft/Phi-4-mini-instruct
Qwen3 4B Instruct 2507	Qwen	4B	Qwen/Qwen3-4B-Instruct-2507
Gemma 3 4B it	Google	4.3B	unsloth/gemma-3-4b-it
Qwen3.5 4B	Qwen	4.7B	Qwen/Qwen3.5-4B
Qwen3 8B	Qwen	8.2B	Qwen/Qwen3-8B

- **Same settings for all:** the model's own chat template, deterministic generation (no random sampling, so the same prompt gives the same answer), at most 96 new tokens, "thinking" mode turned off where the model has one. A leftover thinking block is removed from the answer before scoring.
- **Precision:** 16-bit floats first; if the model then produces gibberish on a test question (this can happen on some models), the run retries in bfloat16, and stops with an error if neither works.
- **Hardware:** Kaggle with 2 Tesla T4 GPUs, one GPU per model by default. Python 3.13, PyTorch 2.11, Transformers 5.20, fastembed 0.9 (versions recorded in `benchmark-cosmos/meta.json` for the second pass).
- **Two passes:** Qwen3 4B and Qwen3 8B ran in a second pass with the same settings. Their full .md prompt (about 5,000 tokens) did not fit in GPU memory in one go, so long prompts were read in pieces of 1,024 tokens. This is the same computation, with less memory.

5. How answers were scored

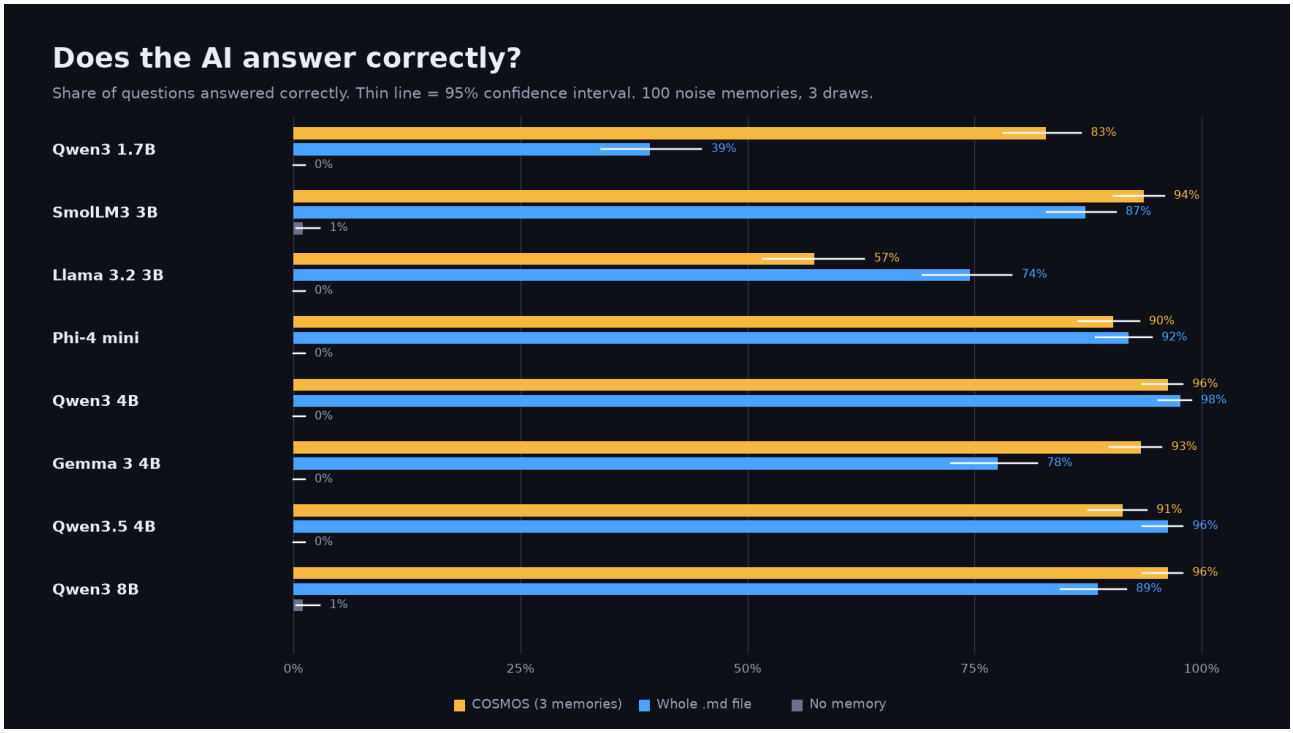
- **Question with an answer:** correct if the answer contains every expected keyword group (accents, case and punctuation ignored) and does not say "I don't know". Numbers must match as whole numbers: "4" does not match inside "14", and "4 500" equals "4500".
- **Question with no answer:** correct only if the model says it does not know (a list of phrases such as "I don't know", "no information", "not mentioned", "not in memory"). A made-up answer counts as wrong.
- **Headline accuracy** counts only questions with an answer (298 per model and setup). The no-answer questions are reported separately as "abstention".
- **Scoring is by keyword, not by a person or a judge model.** It is strict and consistent, but can mark a good answer wrong if it uses other words, and a bad answer right if it happens to include the keyword.

6. Statistics and definitions

Measure	Definition
Accuracy	Correct answers divided by questions asked, with a 95% Wilson confidence interval (the range shown in brackets and as the thin line on the bars).
Verdict (wins / ties / loses)	COSMOS vs full .md on the same model. The gap is called significant only if it is larger than the square root of (half-width of the COSMOS interval squared + half-width of the full .md interval squared). Otherwise: tie.
Tokens	Size of the whole prompt, counted with each model's own tokenizer, averaged over all questions.
Time per answer	Seconds to generate the answer on the Tesla T4, averaged over questions. Includes reading the prompt.
Recall@k	Share of questions where the right memory is among the k first search results (k = 1, 3, 5).
MRR	Mean of 1 divided by the rank of the right memory (1 = always first).
Updates	For an update question: share where the new memory ranks above the old one it replaces.
Abstention (search only)	Best accuracy obtainable by saying "nothing found" when the best match scores below a threshold. The threshold is chosen on the test itself (see limits).
Automatic category	Share of the 30 hand-written memories that COSMOS puts in the category written in the dataset.
Search time	Median time of one search, measured on the Kaggle CPU.
Mean and spread	Search-only numbers are averaged over the random draws (5 draws for part A); the spread is the standard deviation between draws.

7. Results

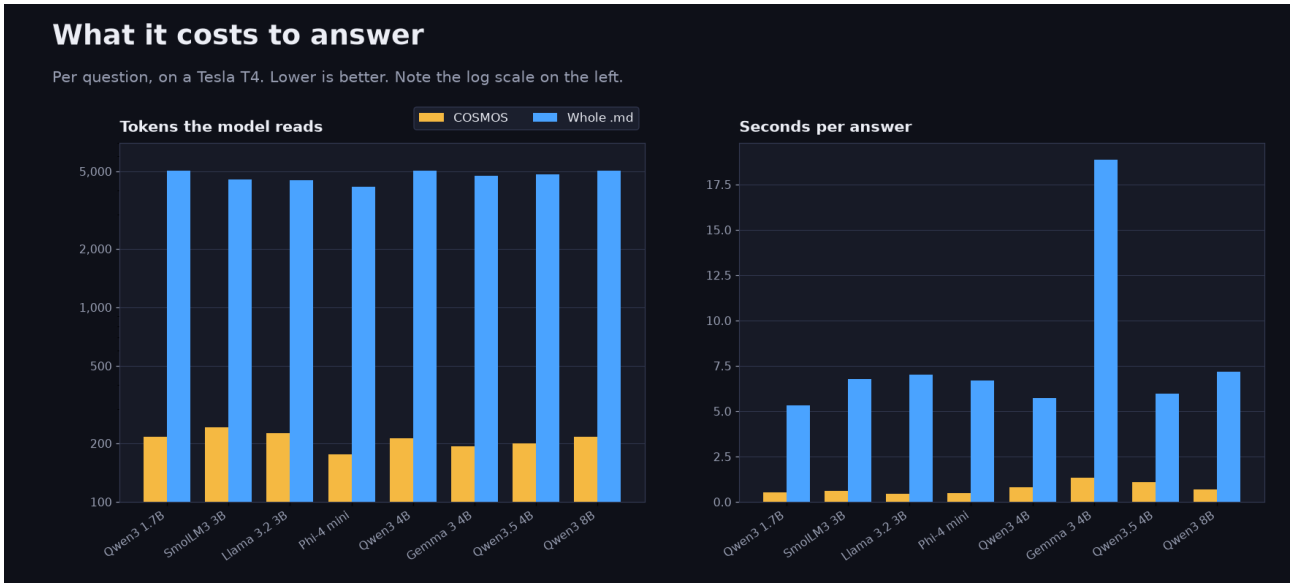
7.1 Answers with an AI (part B): 100 noise memories, 3 draws



Accuracy per model with 95% confidence intervals (same data as the table below).

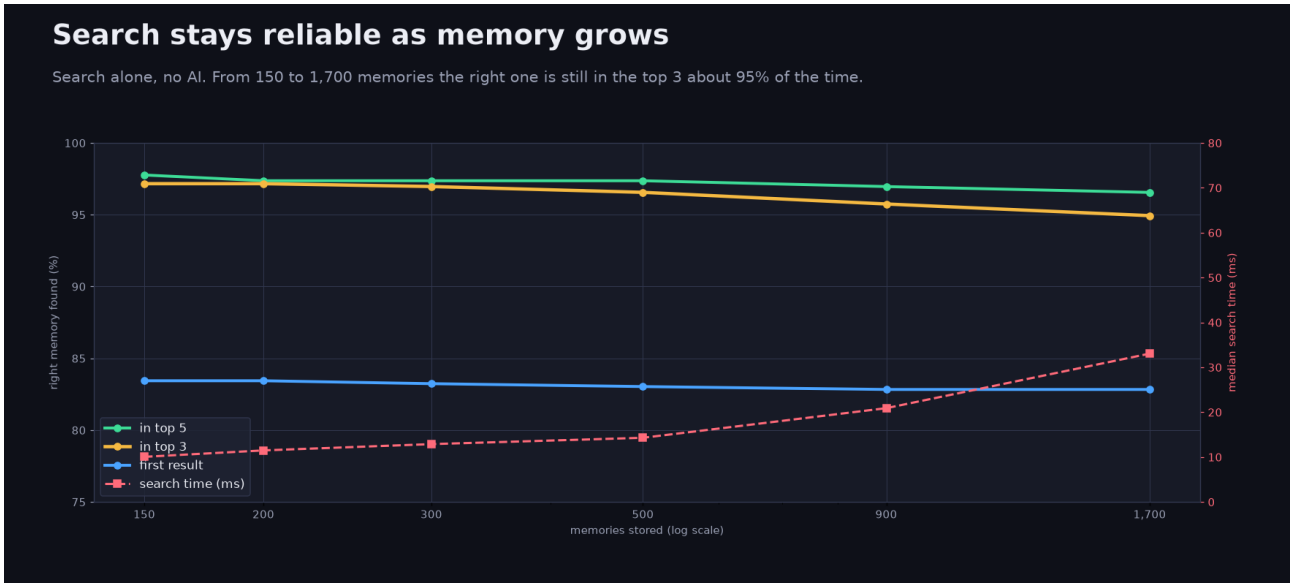
Model	COSMOS	Full .md	No memory	Verdict	Tokens (C / full)	Seconds (C / full)
Qwen3 1.7B	83% [78-87]	39% [34-45]	0%	COSMOS	216 / 5,060	0.5 / 5.3
SmolLM3 3B	94% [90-96]	87% [83-91]	1%	COSMOS	243 / 4,530	0.6 / 6.8
Llama 3.2 3B	57% [52-63]	74% [69-79]	0%	Full .md	227 / 4,514	0.5 / 7.0
Phi-4 mini	90% [86-93]	92% [88-95]	0%	Tie	176 / 4,171	0.5 / 6.7
Qwen3 4B	96% [94-98]	98% [95-99]	0%	Tie	212 / 5,056	0.8 / 5.7
Gemma 3 4B	93% [90-96]	78% [72-82]	0%	COSMOS	193 / 4,752	1.3 / 18.8
Qwen3.5 4B	91% [88-94]	96% [94-98]	0%	Full .md	200 / 4,815	1.1 / 6.0
Qwen3 8B	96% [94-98]	89% [84-92]	1%	COSMOS	216 / 5,060	0.7 / 7.2

C = COSMOS. COSMOS searched with MiniLM during this run (Polaris came later). Averages: about 210 tokens with COSMOS against about 4,750 with the full file (23 times fewer), and about 0.75 s against 7.9 s per answer (11 times faster). Source: `benchmark-cosmos/SUMMARY.md`.



Tokens read and seconds per answer, per model.

7.2 Search alone (part A): memory growing from 150 to 1,700 entries, 5 draws



Search quality and speed as the memory grows.

Memories	Right in top 1	Right in top 3	Right in top 5	MRR	Search time
150	83.4%	97.2%	97.8%	0.903	10 ms
300	83.2%	97.0%	97.4%	0.899	13 ms
500	83.0%	96.6%	97.4%	0.897	14 ms
900	82.8%	95.7%	97.0%	0.894	21 ms
1,700	82.8%	94.9%	96.6%	0.891	33 ms

Search alone used the earlier search model (MiniLM). Other measured values: newest memory ranked above the old one 70% of the time; "no answer" detected 85% of the time; automatic category right 21 to 39%.

8. Polaris, the search model (part C)

Polaris is `intfloat/multilingual-e5-small` (118 million parameters, MIT license) fine-tuned on memories written for COSMOS.

8.1 Training

- **Data:** 7,040 memories written for this purpose, in 12 languages (English, Spanish, French, Italian, German, Portuguese, Dutch, Arabic, Chinese, Japanese, Turkish, Polish). Each has three different ways to ask for it, a near-miss memory that does not answer (same person, another fact), and for about half of them a newer version of the fact. The data was written with AI, not collected from real users.
- **Method:** full fine-tuning with a contrastive loss (MultipleNegativesRankingLoss): each question must land closer to its memory than to the near-miss and to the other memories of the batch. 26,029 training examples, 2 epochs, batch of 64, 2 x Tesla T4, about 10 minutes. 6% of the themes were kept out to check the model on memories it never saw.
- **Fair test:** the names and values used by the COSMOS benchmark were banned from the training data, so the benchmark measures memories Polaris never saw.
- **A first attempt failed:** it taught the model that the old version of a fact is a wrong answer. It then ranked updates first every time, but stopped finding old values and rephrased questions (right memory first in 45% of cases). The final Polaris treats both versions as right answers, and COSMOS puts the newest first using the date.

8.2 Comparison

Search alone, no AI, about 200 stored memories, 3 random draws (about 125 questions each). Three search models: MiniLM (COSMOS's previous model), e5-small without fine-tuning, and Polaris.

Search measure	MiniLM	e5-small	Polaris
Right memory ranked 1st	85.6%	87.3%	85.3%
Right memory in the top 3	97.6%	96.3%	98.3%
Ranking quality (MRR)	0.914	0.920	0.915
"Not in memory" recognized	84.6%	84.8%	92.0%
Newest version of a fact ranked 1st	91.7%	91.7%	86.1%

Polaris is not ahead everywhere. It ranks the right memory 1st slightly less often than the untuned model (2 or 3 questions out of 125, within chance), and it ranks the newest version of a fact 1st less often, because it treats old and new versions as equally right. COSMOS fixes that order afterwards using the date.

9. Limits: what these numbers do not say

- **One memory size with AI.** About 200 memories and 3 random draws. Runs with a larger memory stopped because the GPU ran out of memory. The search-only series goes up to 1,700 memories; the AI series does not.
- **Small and mid-size models only** (1.7 to 8 billion parameters). Nothing here says how it behaves with very large models, which read long prompts better.
- **French test data, one prompt.** Another prompt wording could change who wins.
- **Mostly generated data.** 70 of the 100 non-noise memories, all the noise, and about 85 of the 123 questions per draw are generated from templates with made-up people. Real notes are messier. The 30 hand-written memories are few.
- **Keyword scoring** (section 5), no human or judge model.
- **Settings tuned on the test.** The "no answer" threshold and the version-detection setting were chosen on this same benchmark, so every model looks slightly better than it would on new data.
- **Update questions favor COSMOS** because only COSMOS can mark a replaced memory (section 3).
- **Small samples.** About 125 questions per draw. Gaps of 1 to 2 points are within chance. The verdict rule uses confidence intervals, not a full statistical test across models.
- **Synthetic training data for Polaris**, written by an AI. Real-world gains will likely be smaller.
- **Not a win everywhere.** Against the full file COSMOS loses on Llama 3.2 3B (57% vs 74%) and Qwen3.5 4B (91% vs 96%).
- **Weak spots measured:** automatic categorization is right only 21 to 39% of the time; detecting "no answer" without an AI tops out around 85%.
- **Run by the author.** No independent party has repeated it yet.

10. Reproducing the benchmark

Everything is in the project repository (`github.com/DevLiamDb/COSMOS`). The seeds make every draw identical.

Item	Where
Test data (hand-written)	benchmark/dataset.json
Case generator	benchmark/synth.py

Item	Where
Runner (search, prompts, scoring)	benchmark/run.py
Run every model, then report	benchmark/run_all.py, benchmark/report.py
Notebook used on Kaggle	benchmark/kaggle_notebook.ipynb
Raw results per model	benchmark-cosmos/raw/*.json
Every answer of every model	benchmark-cosmos/tables/toutes_les_reponses.csv
Machine, versions, settings	benchmark-cosmos/meta.json
Charts and tables	benchmark-cosmos/report.html, charts/, tables/

```
# search only, no AI
python -m benchmark.run --scales 50,100,200,400,800,1600 --seeds 0,1,2,3,4
# with one model
python -m benchmark.run --llm Qwen/Qwen3-8B --scales 100 --seeds 0,1,2
```

The numbers will differ slightly on other hardware (speed above all). Accuracy should match closely because generation is deterministic, but small differences between library versions are possible.